

EASY

1)

Algebra relacional

$$\pi_{\text{booking_date, nights}} (\sigma_{\text{guest_id}=1183}(\text{Bookings}))$$

Calculo relacional

$$\{b.\text{booking_date}, b.\text{nights} \mid \text{booking}(b) \wedge b.\text{guest_id}=1183\}$$

2)

Algebra relacional

$$\tau_{\text{arrival_time}} \downarrow (\pi_{\text{arrival_time, first_name, last_name}} (\sigma_{\text{booking_date}='2016-11-05' \wedge \text{booking.guest_id}=\text{guest.id}} (\text{bookings} \times \text{guest})))$$

Calculo Relacional

$$\{b.\text{arrival_time}, g.\text{first_name}, g.\text{last_name} \mid b.\text{booking}, g.\text{guest} \mid b.\text{booking_date}='2016-11-05' \wedge b.\text{guest_id}=g.\text{id}\}$$

3)

Algebra relacional

$$\pi_{\text{booking_id, room_type, occupants, amount}} (\sigma_{\text{booking_id} \in \{5152, 5165, 5154, 5295\} \wedge \text{booking.occupants}=\text{rate.occupancy} \wedge \text{booking.room_type}=\text{requested}=\text{rate.room_type}} (\text{bookings} \times \text{rate}))$$

calculo relacional

$$\left\{ \begin{array}{l} b.booking_id, b.room_type_requested, b.occupants, r.amounts | b.booking, r.rate | \\ b.booking_id \in \{5112, 5165, 5154, 5295\} \wedge b.occupants = r.occupancy \wedge \\ b.room_type_requested = r.room_type \end{array} \right\}$$

4)

Algebra relacional

$$\pi_{first_name, last_name, address} (\sigma_{room_no=101 \wedge booking_date='2016-12-03' \wedge booking.guest_id = guest.id} (booking \times guest))$$

calculo relacional

$$\left\{ g.first_name, g.last_name, g.address | b:booking, g:guest | b.room_no=101 \wedge b.booking_date='2016-12-03' \wedge b.guest_id=g.id \right\}$$

5) Algebra relacional

$$\pi_{guest_id, count(nights), sum(nights)} (\sigma_{guest_id \in \{1185, 1270\}} (booking))$$

calculo relacional

$$\left\{ b.guest_id, count(b.nights), sum(b.nights) | b:booking | b.guest_id \in \{1185, 1270\} \right\}$$